Implementation Plan: Smart Content Summarizer

1. Project Setup

1. Open Google Opal and access the visual editor.
2. Create or open the existing SummaSmart application.
3. Save changes after configuring each step.

2. Workflow Design

Implement the following five steps in order.

Step 1: User Instructions

- Collect summary length, format, language, tone, and detail requirements.
- Pass the instructions to the summary generation step.

Step 2: Get Content Source

- Ask the user for an article URL.
- Preserve the URL as entered.
- Pass the URL to the retrieval step.

Step 3: Fetch content

- Connect the Get Webpage tool.
- Pass the Get Content Source output as the tool's URL input.
- Return retrieved webpage text to the next step.
- Report retrieval failures clearly.

Step 4: Generate Content Summary

- Provide the retrieved article as one context variable.
- Provide user instructions as a second context variable.
- Generate a summary using the article as the factual source.
- Respect the requested output format.

Step 5: Render Summary Webpage

- Display the user's instructions.
- Display the original source URL as a clickable Markdown link.
- Display the generated summary.
- Display the original retrieved webpage text.
- Use Markdown headings and readable spacing.

3. Data Flow

User Instructions → Generate Content Summary

Get Content Source → Fetch content → Generate Content Summary

User Instructions + Source URL + Retrieved Content + Generated Summary → Render Summary Webpage

4. Configuration Principles

- Keep Objective and Output Format instructions clear and separate when the editor provides separate fields.
- Verify context bindings against actual node names and paths in the Opal editor.
- Do not assume that text pasted into a prompt automatically configures a tool connection.
- Confirm the retrieval tool's URL input is actually connected.
- Avoid changing working prompts without a specific reason.

5. Testing Plan

1. Test with `https://example.com`.
2. Check the generated summary against the retrieved content.
3. Request a fixed number of bullet points.
4. Test a second publicly accessible article.
5. Check the behavior when a URL cannot be retrieved.
6. Verify the final webpage displays all required sections.
7. Record actual results, including errors and limitations.

6. GitHub Documentation

Create a public repository named `smart-content-summarizer-opal`.

Add:

- `prompts/user_instructions_prompt.txt`
- `prompts/fetch_content_prompt.txt`
- `prompts/generate_summary_prompt.txt`
- `prompts/render_summary_prompt.txt`
- `concept_guide.md`
- `requirement_document.md`
- `implementation_plan.md`
- `testing_plan.md`
- `README.md`

7. Demonstration Video Plan

Record the workflow canvas, node configurations, connected retrieval tool, test inputs, and final output. Show a successful example and explain the workflow's limitations honestly.

8. Completion Checklist

- Workflow saved.
- All five steps configured.
- Retrieval tool connection verified.
- Summary context variables verified.
- Final Markdown output verified.
- Example test recorded.
- Second article test recorded.
- Error-handling test recorded.
- GitHub files uploaded.
- Public repository checked.
- Explanation video recorded and reviewed.